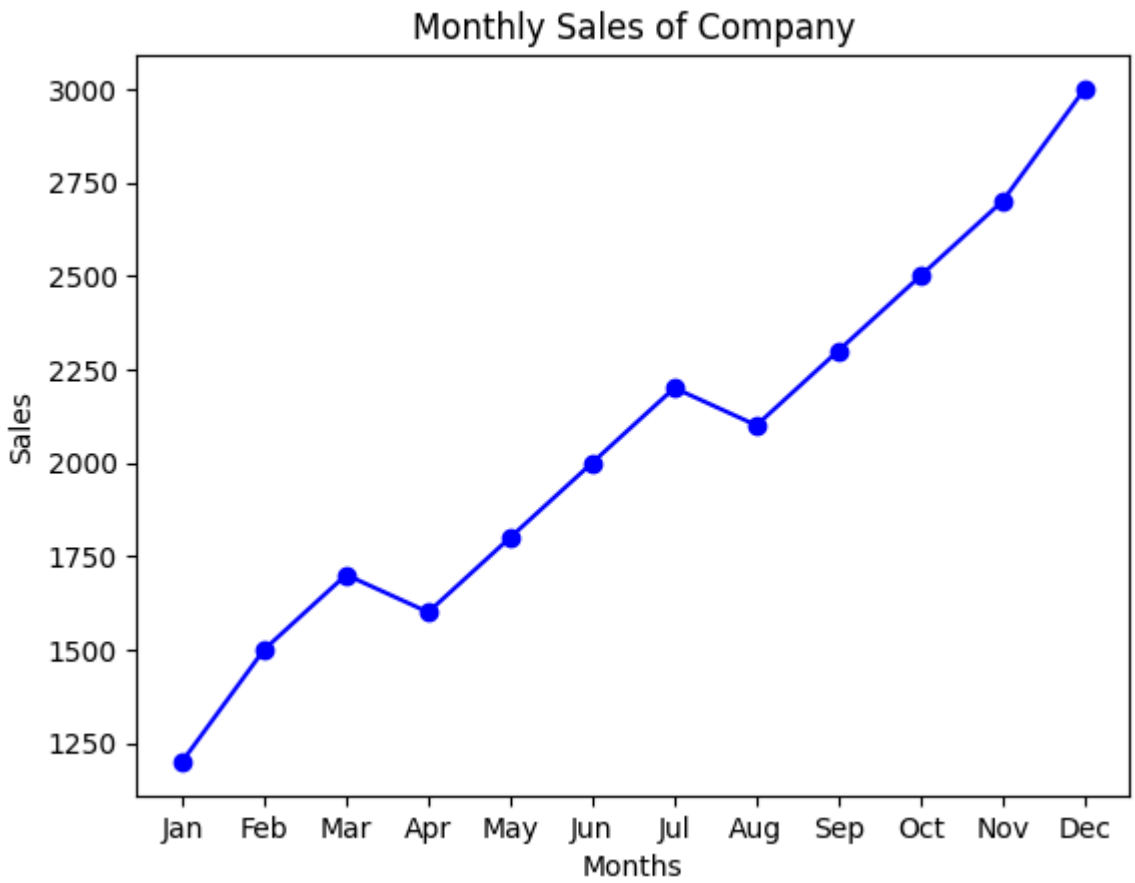
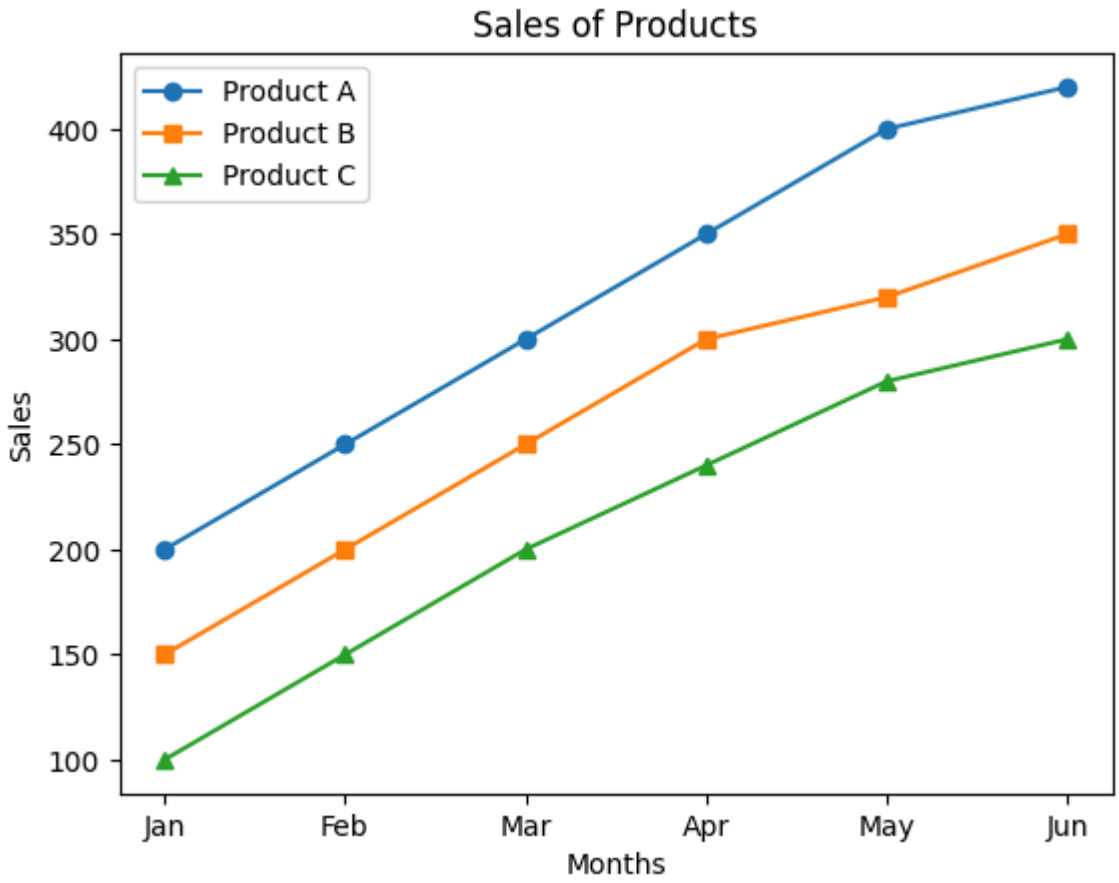
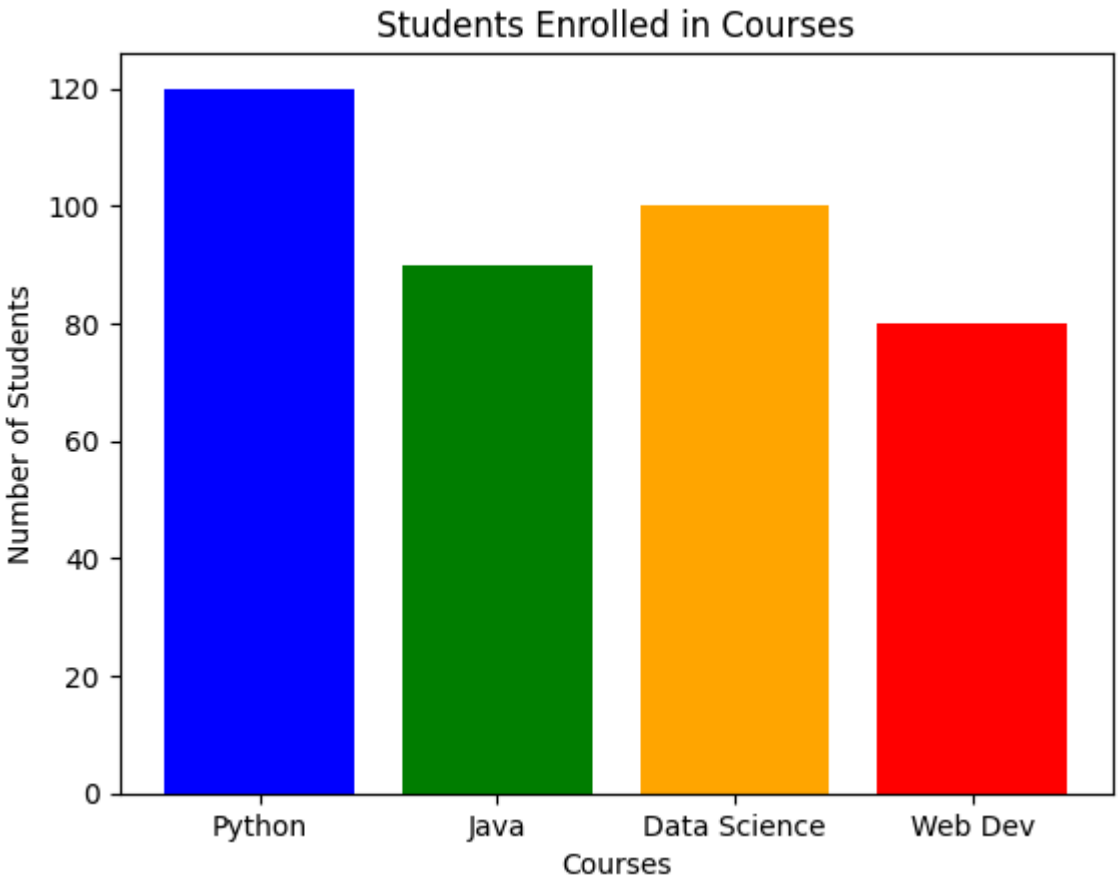
In [1]:
```
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec
sales = [1200,1500,1700,1600,1800,2000,2200,2100,2300,2500,2700,3000]

plt.plot(months, sales, marker='o', color='blue')
plt.title("Monthly Sales of Company")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```



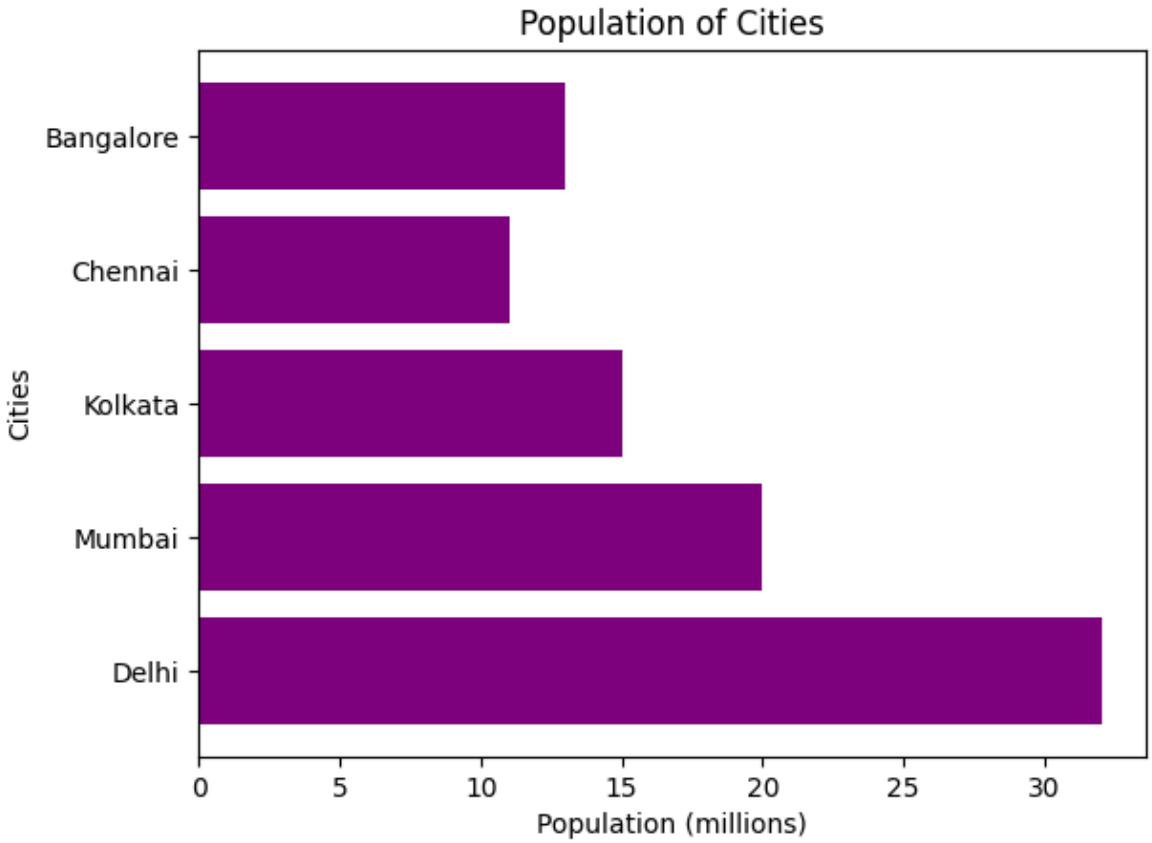
In [2]:
```
months = ["Jan","Feb","Mar","Apr","May","Jun"]

productA = [200,250,300,350,400,420]
productB = [150,200,250,300,320,350]
productC = [100,150,200,240,280,300]

plt.plot(months, productA, marker='o', label="Product A")
plt.plot(months, productB, marker='s', label="Product B")
plt.plot(months, productC, marker='^', label="Product C")

plt.title("Sales of Products")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.legend()
plt.show()
```



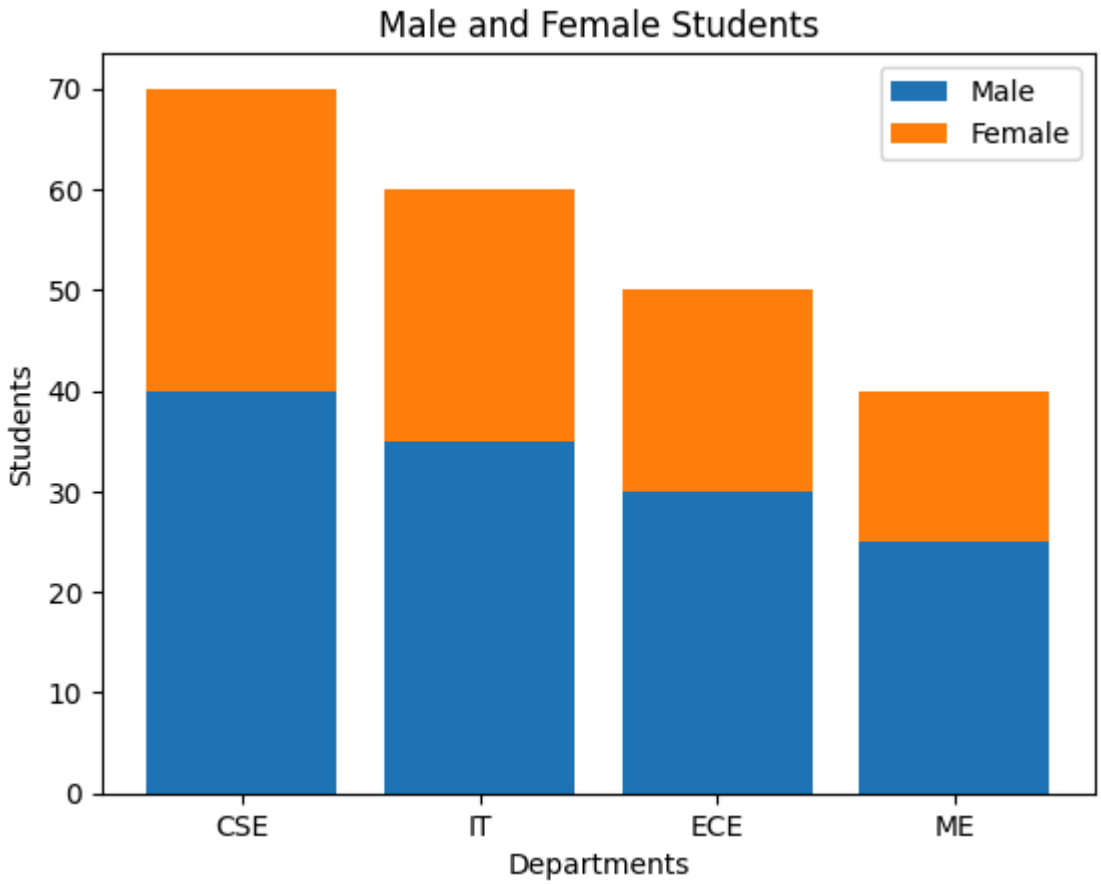
In [3]:
```
courses = ["Python","Java","Data Science","Web Dev"]
students = [120,90,100,80]

plt.bar(courses, students, color=['blue','green','orange','red'])
plt.title("Students Enrolled in Courses")
plt.xlabel("Courses")
plt.ylabel("Number of Students")
plt.show()
```



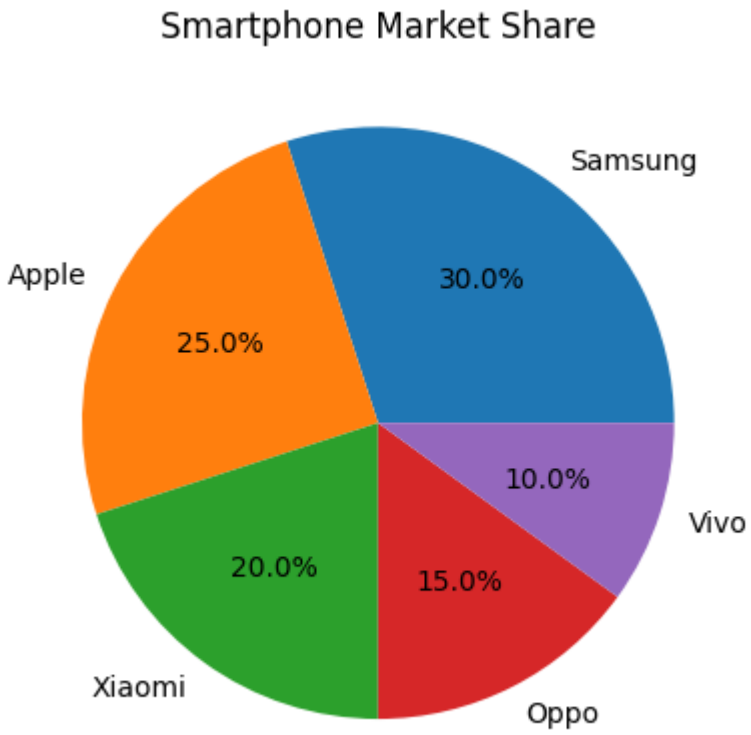
In [4]:
```
cities = ["Delhi","Mumbai","Kolkata","Chennai","Bangalore"]
population = [32,20,15,11,13]

plt.barh(cities, population, color='purple')
plt.title("Population of Cities")
plt.xlabel("Population (millions)")
plt.ylabel("Cities")
plt.show()
```



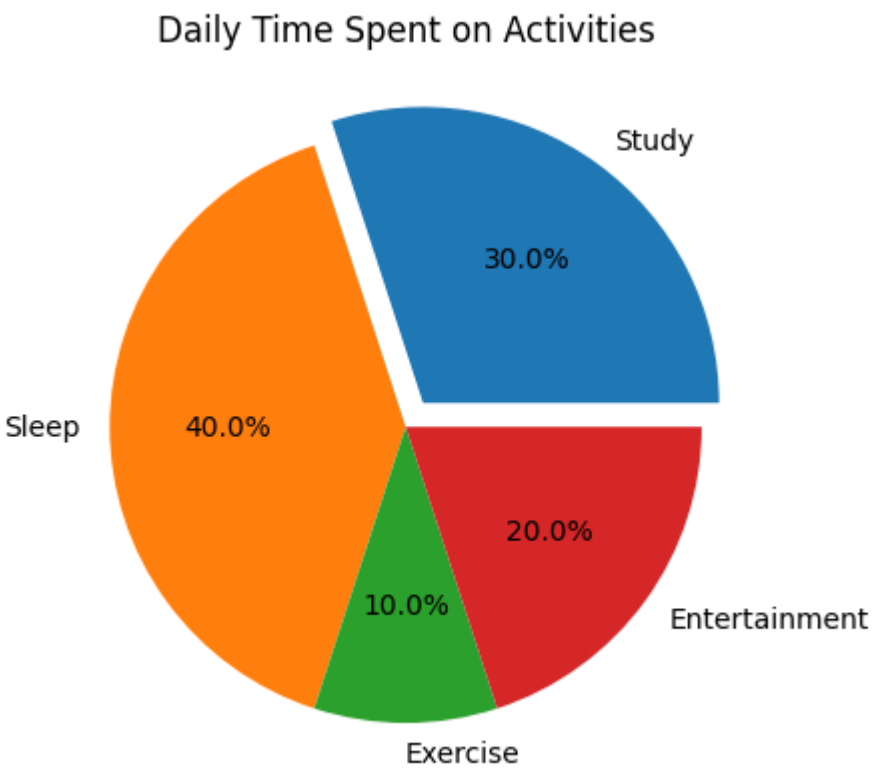
In [5]:
```
departments = ["CSE","IT","ECE","ME"]
male = [40,35,30,25]
female = [30,25,20,15]

plt.bar(departments, male, label="Male")
plt.bar(departments, female, bottom=male, label="Female")

plt.title("Male and Female Students")
plt.xlabel("Departments")
plt.ylabel("Students")
plt.legend()
plt.show()
```



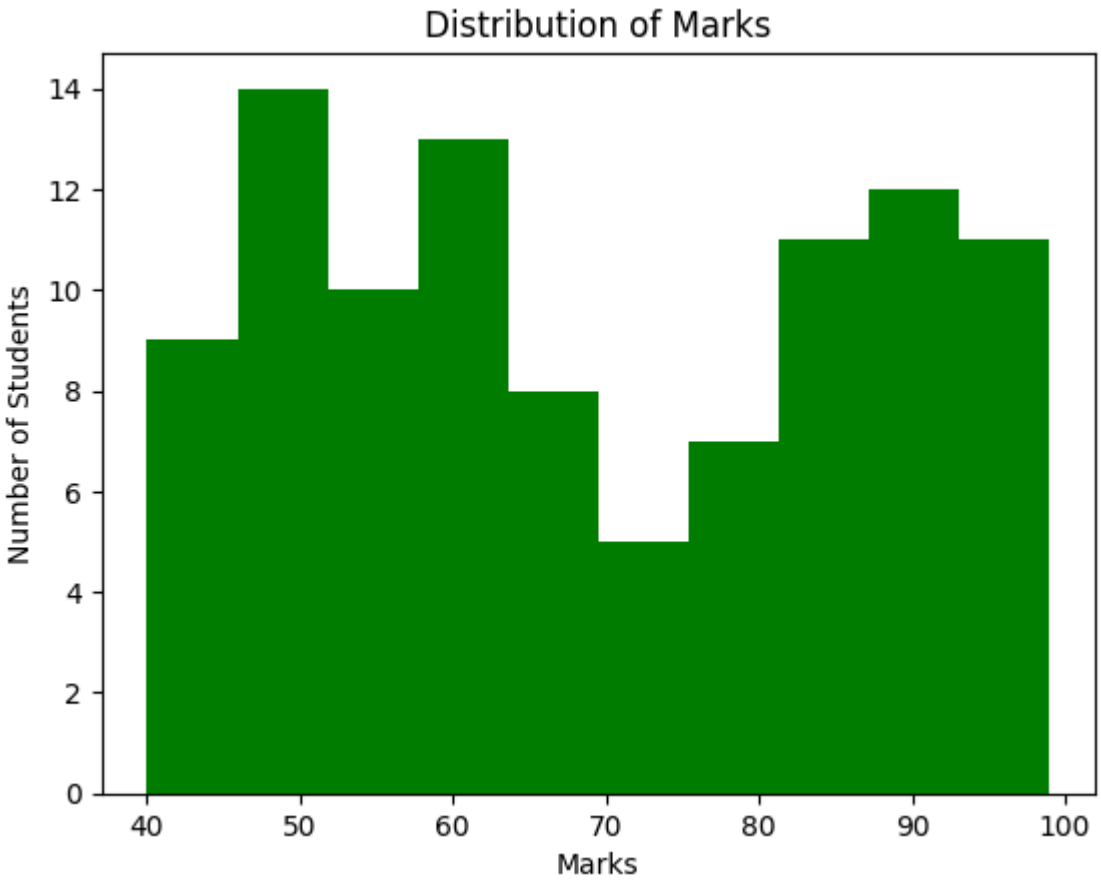
In [6]:
```
brands = ["Samsung","Apple","Xiaomi","Oppo","Vivo"]
share = [30,25,20,15,10]

plt.pie(share, labels=brands, autopct='%1.1f%%')
plt.title("Smartphone Market Share")
plt.show()
```



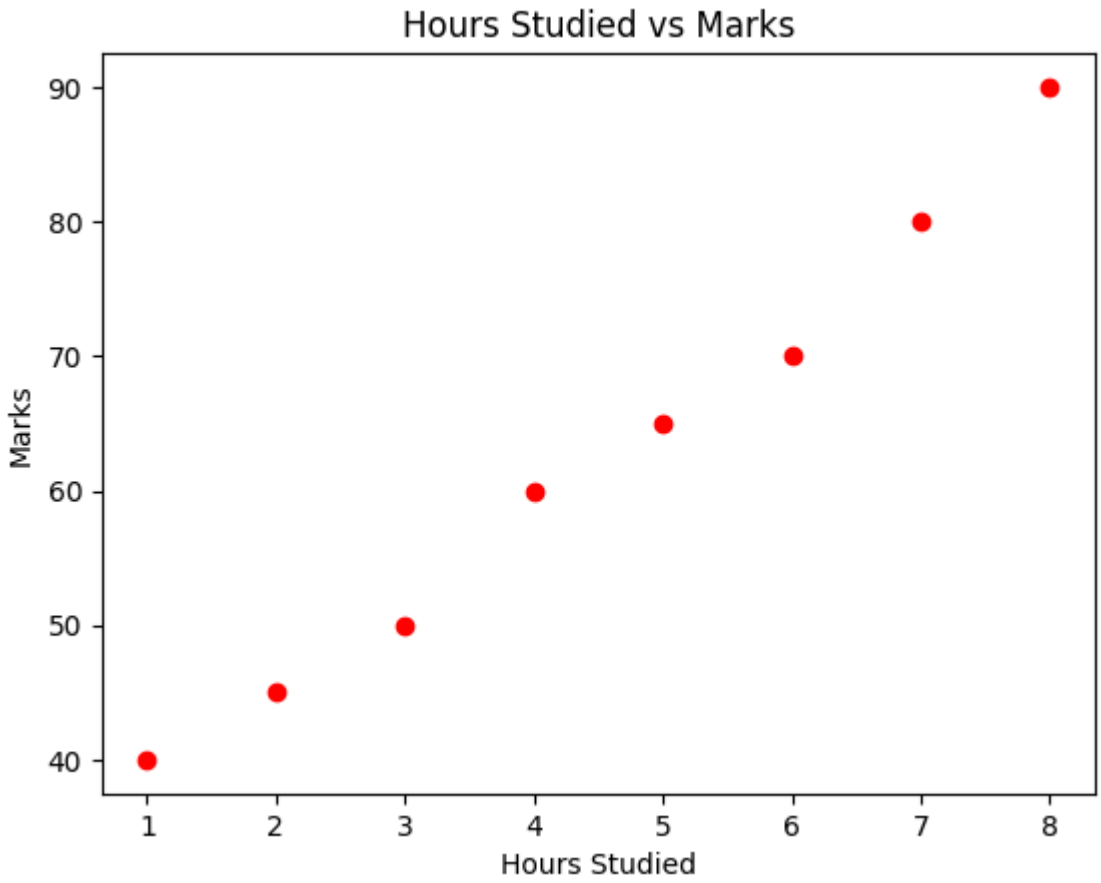
In [7]:
```
activities = ["Study","Sleep","Exercise","Entertainment"]
hours = [6,8,2,4]
explode = (0.1,0,0,0)

plt.pie(hours, labels=activities, autopct='%1.1f%%', explode=explode)
plt.title("Daily Time Spent on Activities")
plt.show()
```



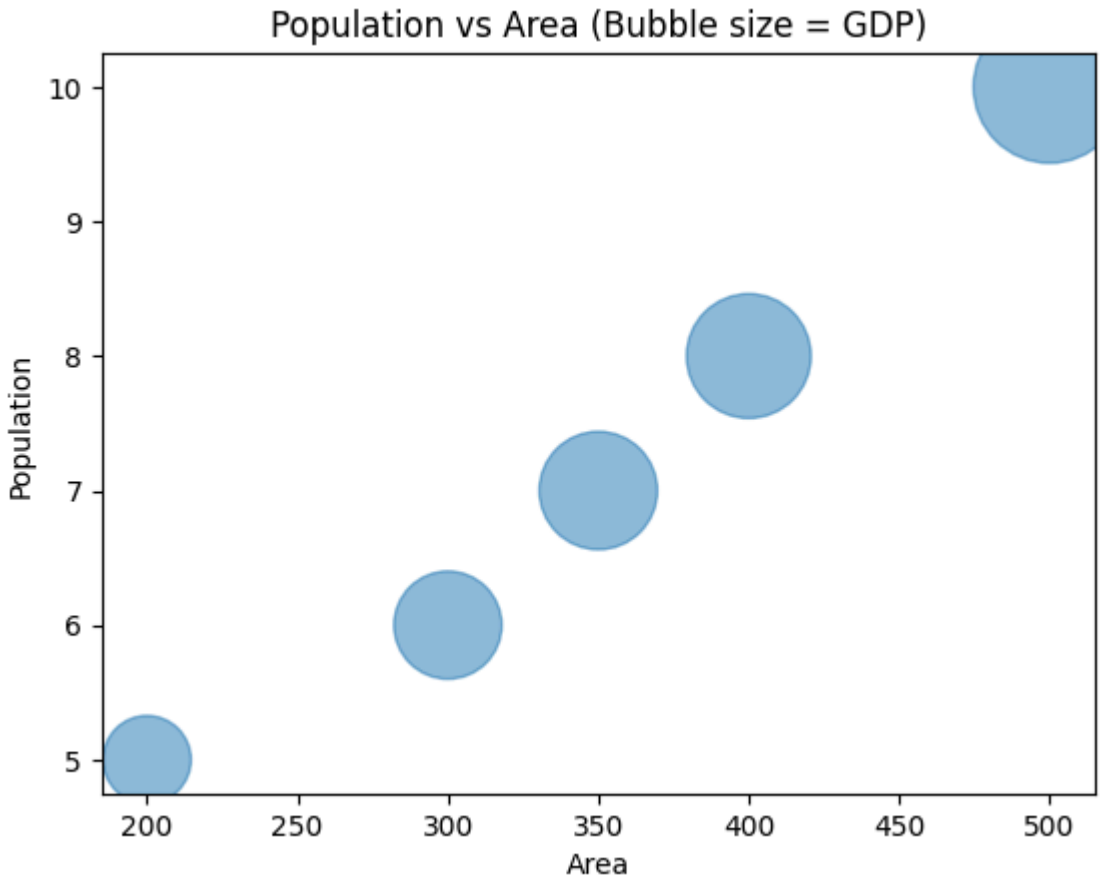
In [8]:
```
import numpy as np

marks = np.random.randint(40,100,100)

plt.hist(marks, bins=10, color='green')
plt.title("Distribution of Marks")
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.show()
```



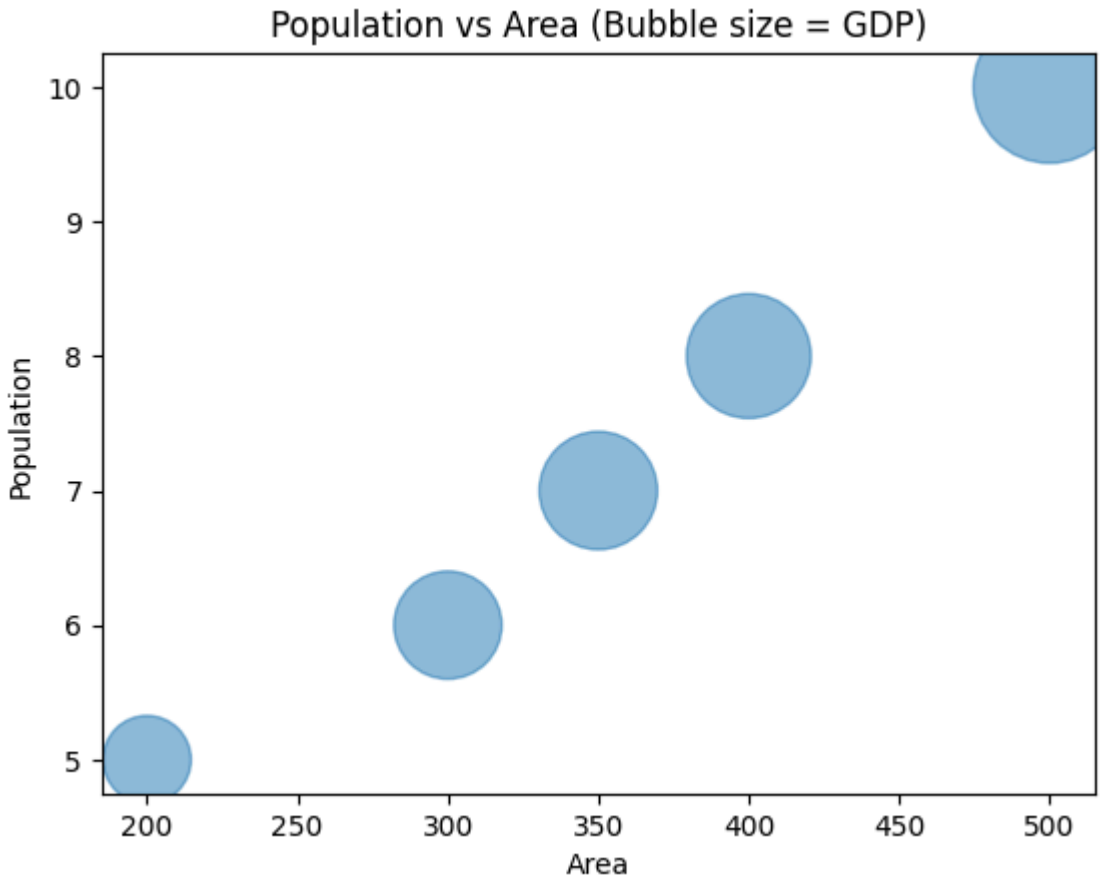
In [9]:
```
hours = [1,2,3,4,5,6,7,8]
marks = [40,45,50,60,65,70,80,90]

plt.scatter(hours, marks, color='red')
plt.title("Hours Studied vs Marks")
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.show()
```



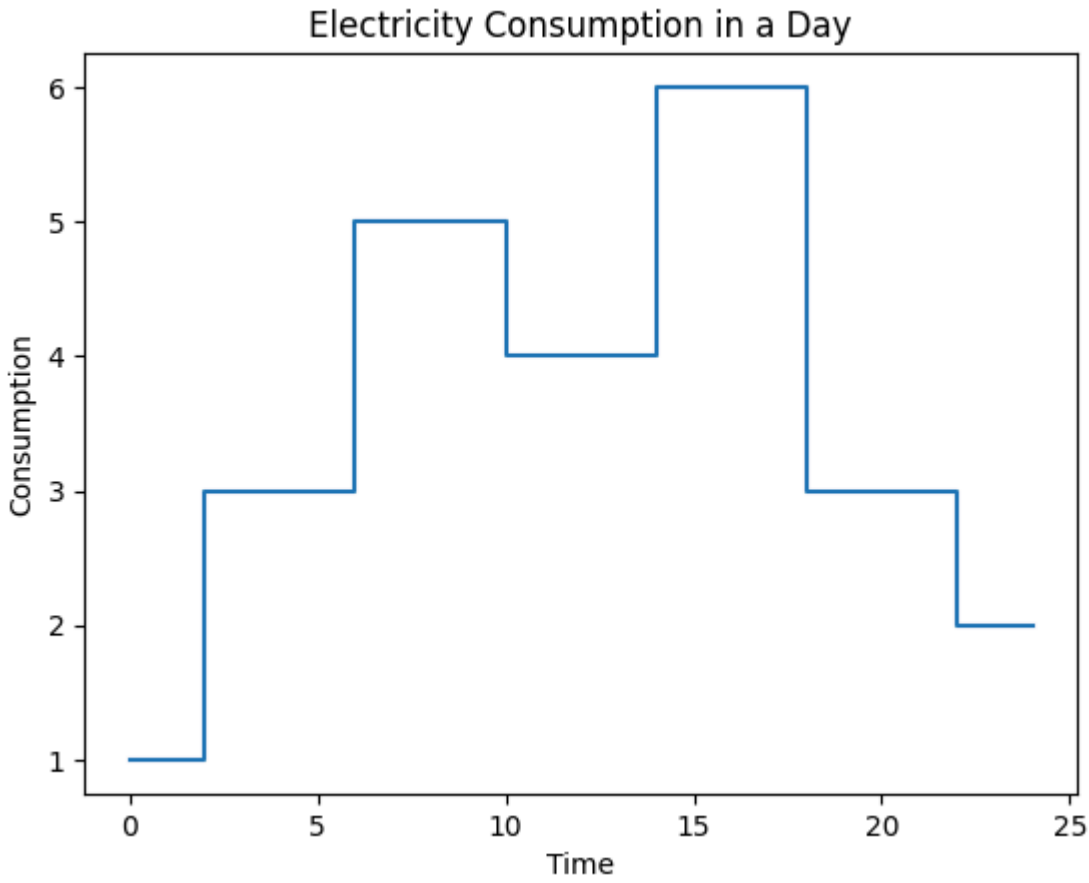
In [10]:
```
cities = ["A","B","C","D","E"]
population = [5,8,10,6,7]
area = [200,400,500,300,350]
gdp = [1000,2000,3000,1500,1800]

plt.scatter(area, population, s=gdp, alpha=0.5)
plt.title("Population vs Area (Bubble size = GDP)")
plt.xlabel("Area")
plt.ylabel("Population")
plt.show()
```



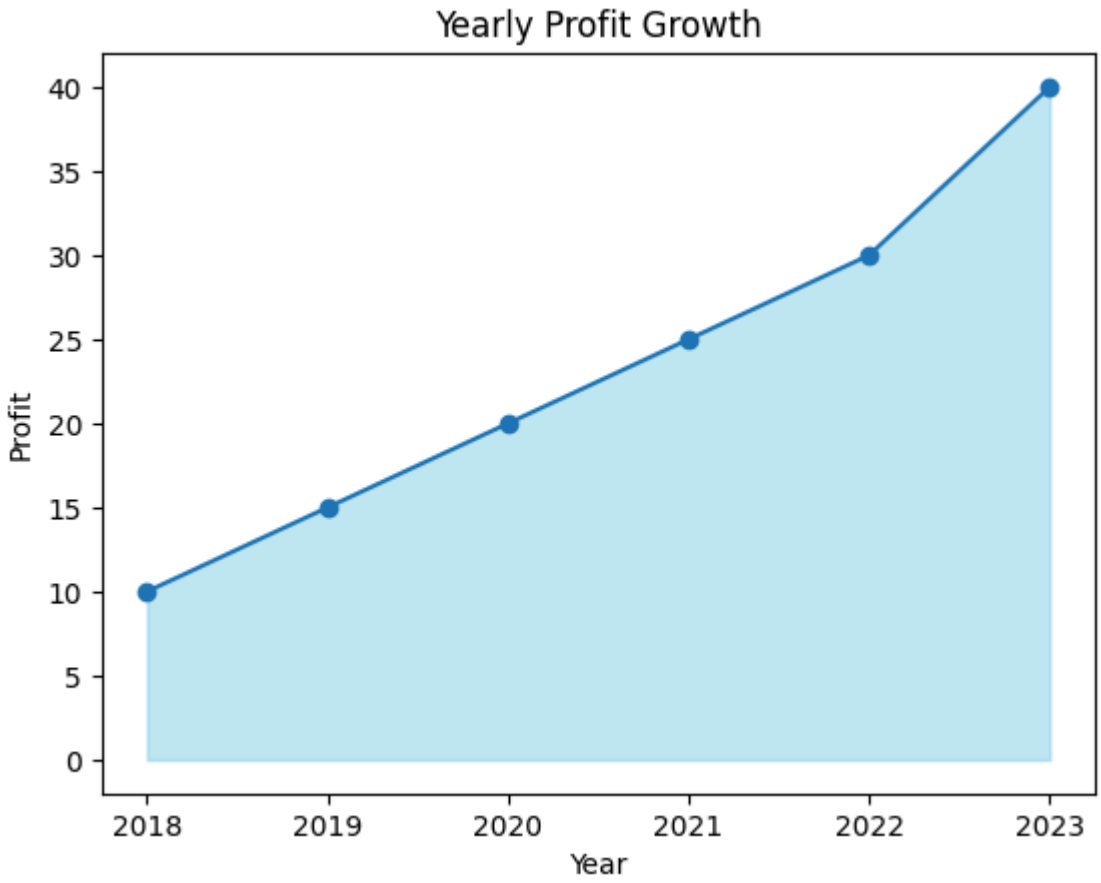
In [11]:
```
cities = ["A","B","C","D","E"]
population = [5,8,10,6,7]
area = [200,400,500,300,350]
gdp = [1000,2000,3000,1500,1800]

plt.scatter(area, population, s=gdp, alpha=0.5)
plt.title("Population vs Area (Bubble size = GDP)")
plt.xlabel("Area")
plt.ylabel("Population")
plt.show()
```



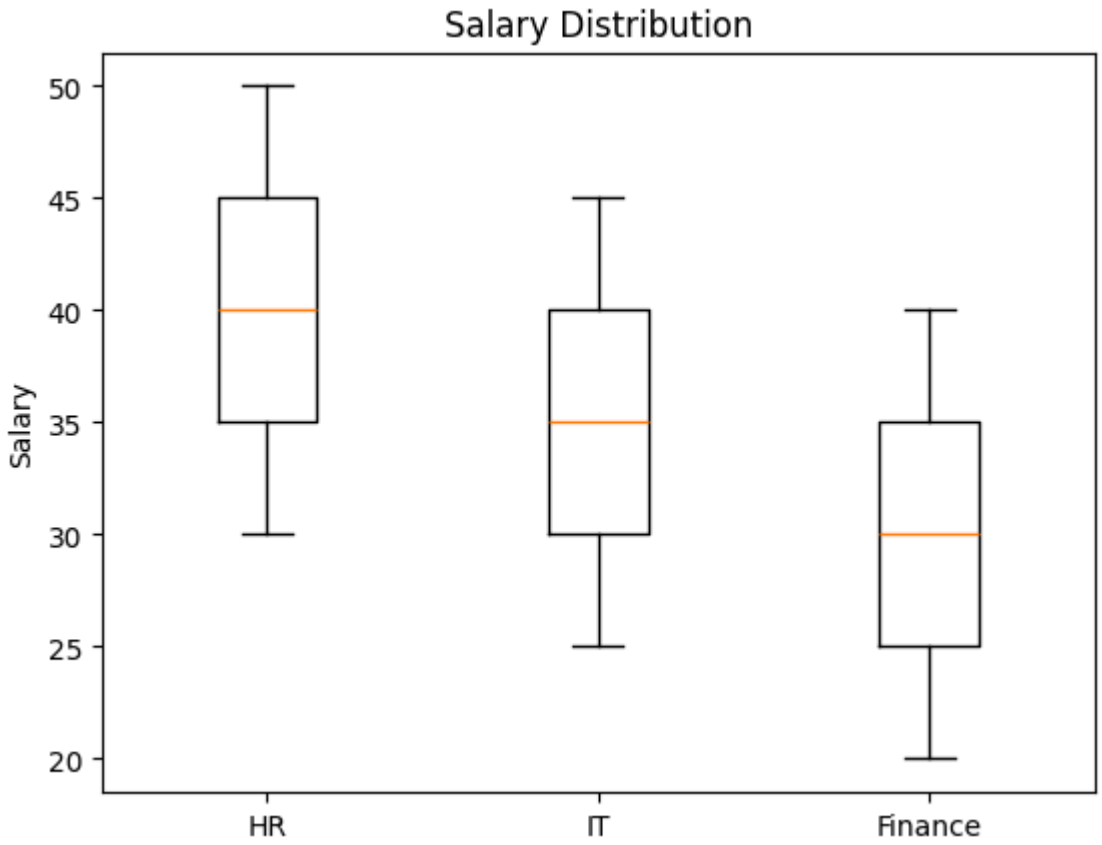
In [12]:
```
time = [0,4,8,12,16,20,24]
consumption = [1,3,5,4,6,3,2]

plt.step(time, consumption, where='mid')
plt.title("Electricity Consumption in a Day")
plt.xlabel("Time")
plt.ylabel("Consumption")
plt.show()
```



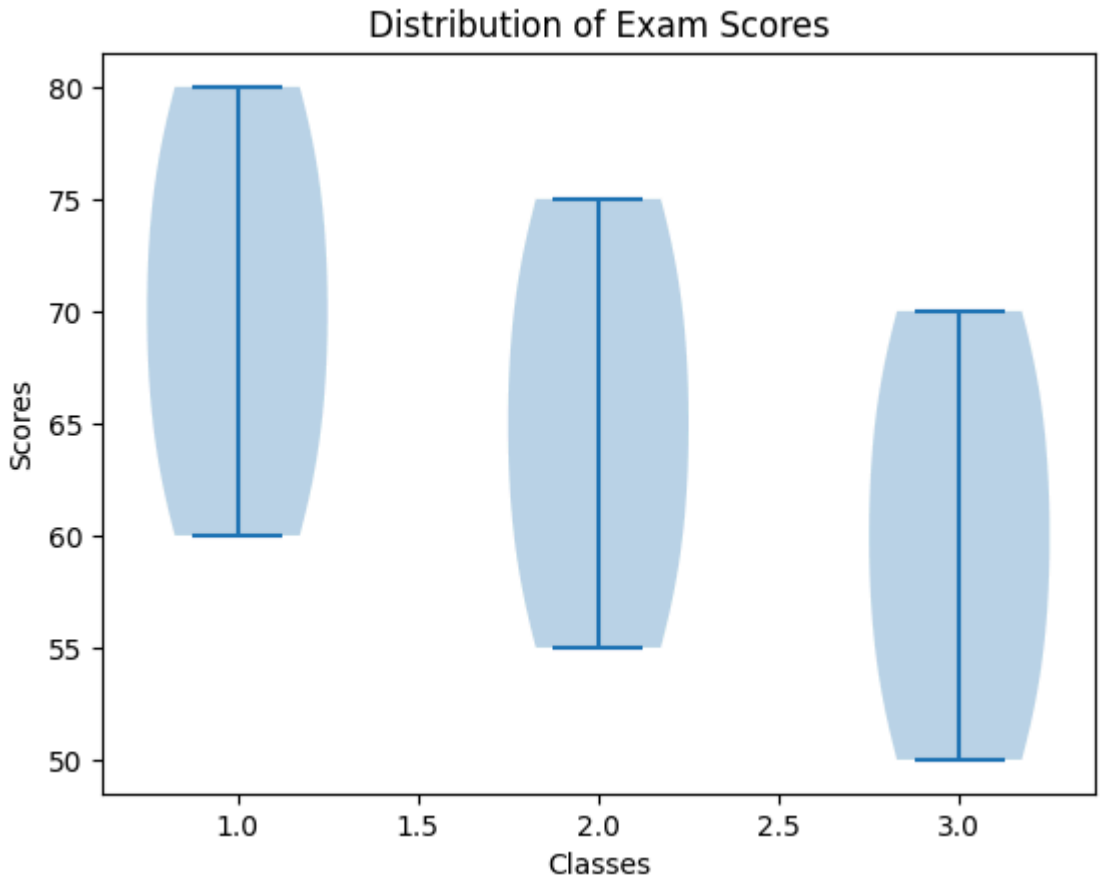
In [13]:
```
years = [2018,2019,2020,2021,2022,2023]
profit = [10,15,20,25,30,40]

plt.fill_between(years, profit, color="skyblue", alpha=0.5)
plt.plot(years, profit, marker='o')
plt.title("Yearly Profit Growth")
plt.xlabel("Year")
plt.ylabel("Profit")
plt.show()
```



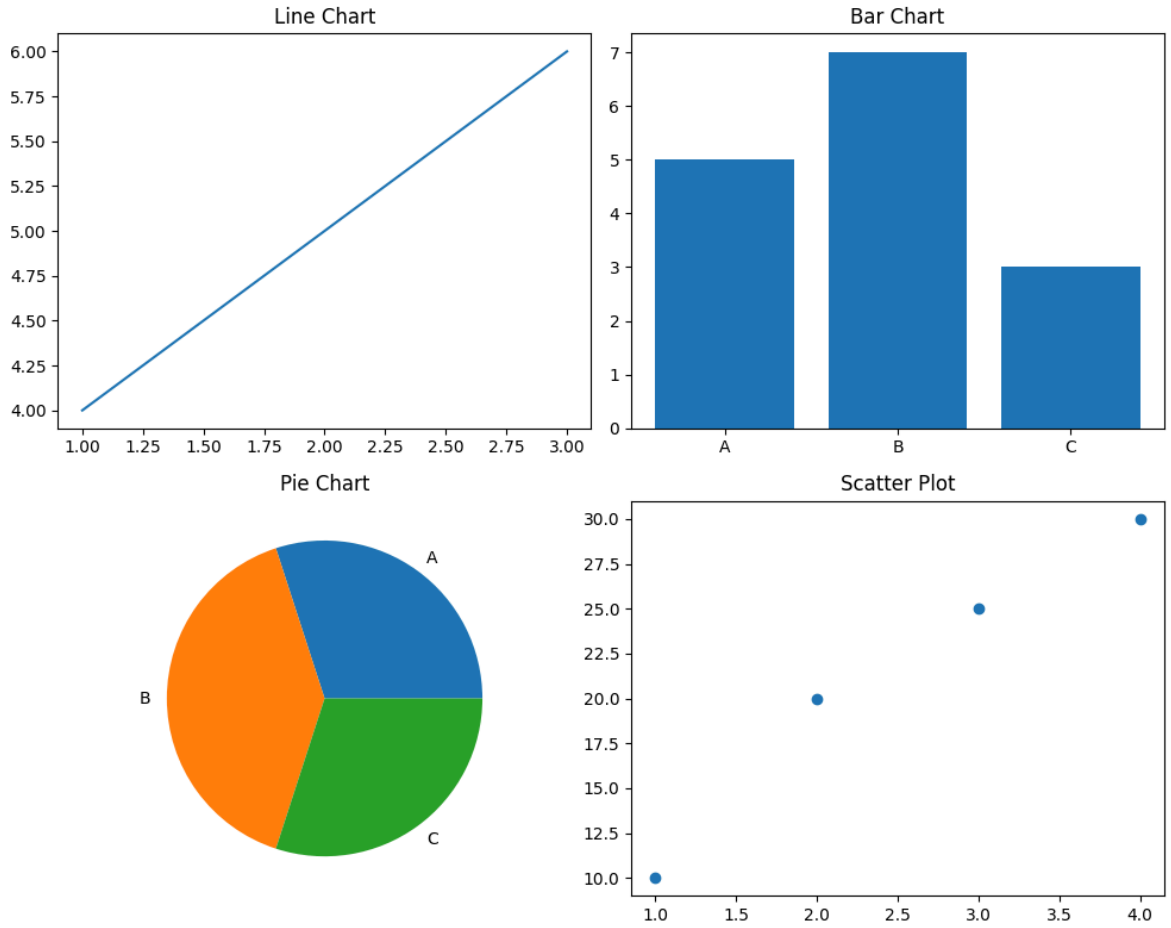
In [15]:
```
dept1 = [30,35,40,45,50]
dept2 = [25,30,35,40,45]
dept3 = [20,25,30,35,40]

data = [dept1, dept2, dept3]

plt.boxplot(data, tick_labels=["HR","IT","Finance"])
plt.title("Salary Distribution")
plt.ylabel("Salary")
plt.show()
```



In [16]:
```
scores = [
    [60,65,70,75,80],
    [55,60,65,70,75],
    [50,55,60,65,70]
]

plt.violinplot(scores)
plt.title("Distribution of Exam Scores")
plt.xlabel("Classes")
plt.ylabel("Scores")
plt.show()
```



In [17]:
```
fig, ax = plt.subplots(2,2, figsize=(10,8))

# Line
ax[0,0].plot([1,2,3],[4,5,6])
ax[0,0].set_title("Line Chart")

# Bar
ax[0,1].bar(["A","B","C"],[5,7,3])
ax[0,1].set_title("Bar Chart")

# Pie
ax[1,0].pie([30,40,30], labels=["A","B","C"])
ax[1,0].set_title("Pie Chart")

# Scatter
ax[1,1].scatter([1,2,3,4],[10,20,25,30])
ax[1,1].set_title("Scatter Plot")

plt.tight_layout()
plt.show()
```


Line Chart
6.00
5.75
5.50
5.25
5.00
4.75
4.50
4.25
4.00
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Bar Chart
7
6
5
4
3
2
1
0
A
B
C
Pie Chart
A
B
C
Scatter Plot
30.0
27.5
25.0
22.5
20.0
17.5
15.0
12.5
10.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0

In [ ]: